



Introduction to Blockchain

CSE-355

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Spring 2026

Lecture 07

Tuesday, Feb 10, 2026

Recap: Money

- **Barter:** Direct exchange of commodities (3-way possible)
- **Commodity**
 - Precious stones, Seashells, Copper, Gold, Tobacco, etc.
 - Need Divisibility, Stability + Scarcity (it has **intrinsic** value!)
- **Representative**
 - Physical notes, certificates, digital (**modern-day stablecoins**)
 - No intrinsic value, but backed by some commodity of intrinsic value
- **Fiat currency**
 - Physical notes as well as in digital format
 - No intrinsic value, nor is it backed by a valuable commodity

USD transitioned from Representative to Fiat money in 1971

Does Fiat always work?

- Recall: Fiat works on trust in the government
 - Not backed by any valuable commodity
- What if the trust in government is low or lost?
 - Sharp and uncontrolled increase in the prices of goods
 - Requires printing/minting even more money
 - Money may quickly become worthless – **hyperinflation**
- Food for thought
 - How do BTC and ETH maintain (or enhance) their value?

Hyperinflation of Germany 1923

The First World War (1914) was very expensive. Germany:

- Suspended the gold-backed currency
- They wanted to conserve gold (for war financing)
- Funded the war by borrowing, instead of imposing taxes
- Believed they would defeat allies and impose war damages
- And then they would be able to pay off the debt
- The German Mark against the USD declined
 - Went from 4.2 to 7.9 per USD between 1914 and 1918
 - This is not so bad. Yet!

Hyperinflation of Germany 1923

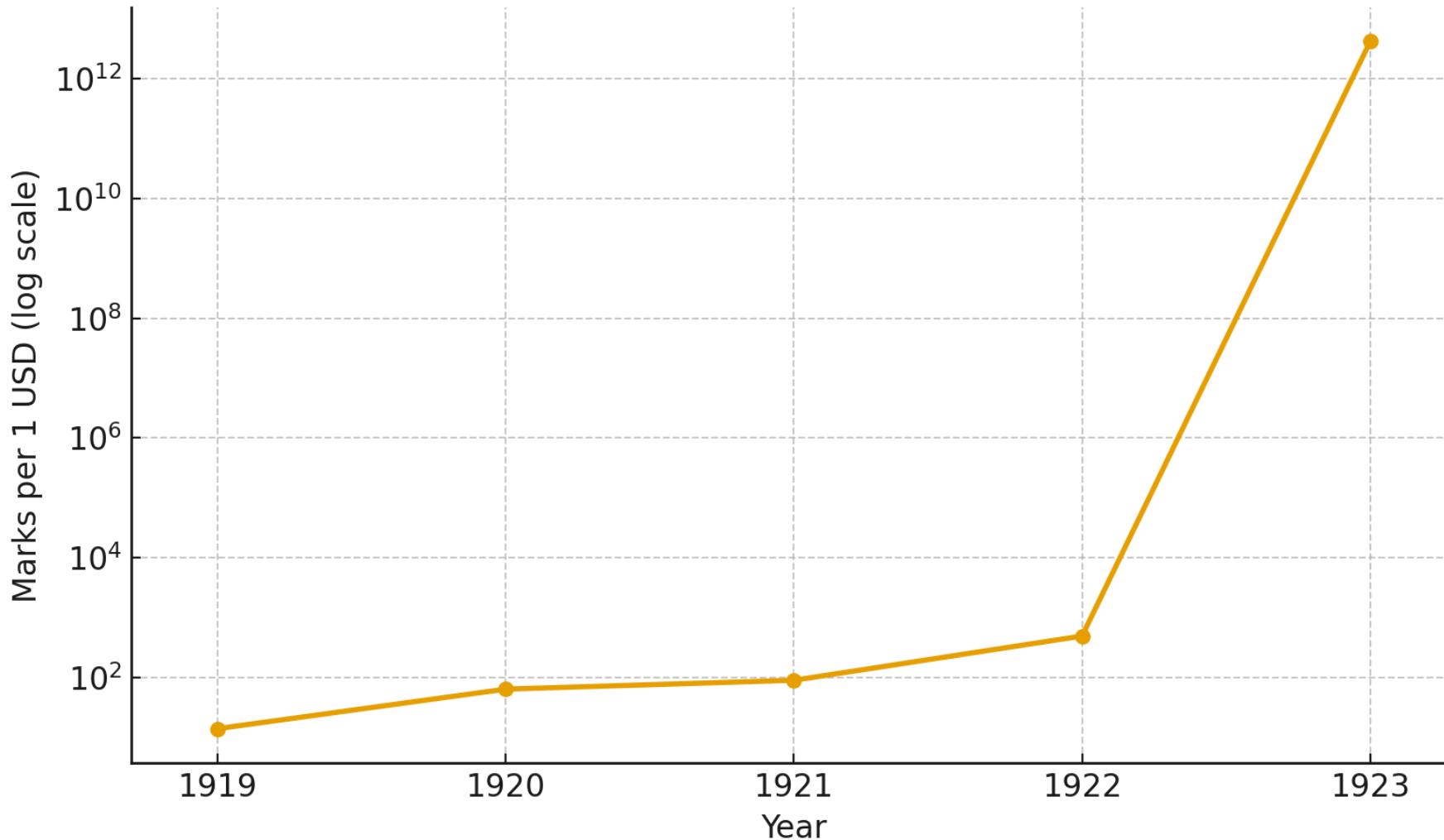
- Germany lost the war (which ended in Nov 1918)
- The German national debt stood at 156B Marks in 1918
 - Printed more money without any economic resources to back it
- Treaty of Versailles (signed: 28 June 1919)
 - Allies demanded 132B Gold Marks (USD 33B) in war reparations
 - Payment required in gold, foreign currency, coal, ships, etc.
 - First payment: 50B Gold-backed Marks in June 1921
 - The value of Mark decreased: 330 to a USD
 - By Dec 1922, the Mark fell to 7400 to a USD

Hyperinflation of Germany 1923

- French and Belgian troops occupied the Ruhr valley
 - Germany's main industrial region, in January 1923
- The occupation was to ensure the payment of reparations
- Germans responded with the policy of passive resistance
 - Workers stopped working, and factories were shut down
 - Industrial productivity decreased significantly
 - Low tax collection
- The government still had to pay workers
- They started printing more and more banknotes

Hyperinflation of Germany 1923

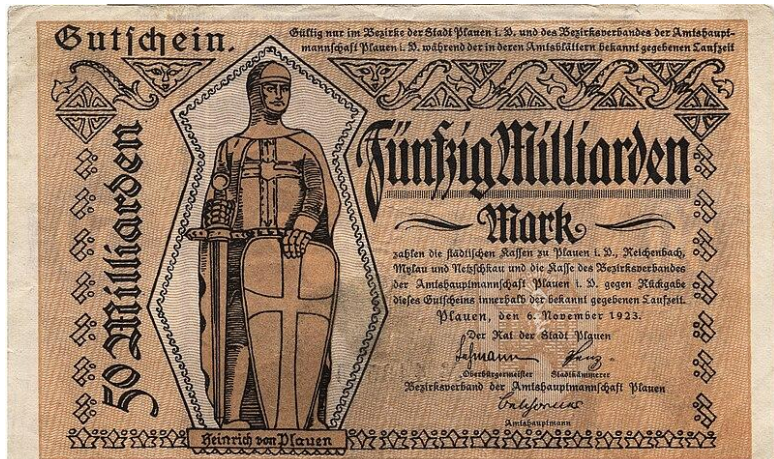
German Mark per US Dollar (1919-1923)



By November 1923, one US dollar was worth 4,210,500,000,000 German marks

Four trillion, two hundred ten billion, five hundred million German marks to a USD

Hyperinflation of Germany 1923



5 trillion marks
Stuttgart, 1923



50 billion (50 Milliarden) marks, Plauen, 1923



50 trillion marks
Eschweiler, 1923

Hyperinflation of Germany 1923



Source: <https://rarehistoricalphotos.com/banknotes-german-hyperinflation-1923/>

Hyperinflation of Germany 1923



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Hyperinflation of Germany 1923



Source: <https://lawliberty.org/monetary-lessons-from-weimar-germany/>

Hyperinflation of Germany 1923



Source: <https://keripeardon.wordpress.com/2013/06/01/wheelbarrows-of-money-and-the-weimar-republic/>

Hyperinflation of Germany 1923

“My father was a lawyer,” says Walter Levy, an internationally known German-born oil consultant in New York, “and he had taken out an insurance policy in 1903, and every month he had made the payments faithfully. It was a 20-year policy, and when it came due, ...

he cashed it in and bought a single loaf of bread.”

*The Berlin publisher Leopold Ullstein wrote that an American visitor **tipped their cook one dollar**. The family convened, and it was decided that ...*

*A **trust fund** should be set up in a Berlin bank with the cook as beneficiary, the **bank to administer and invest** the dollar.*

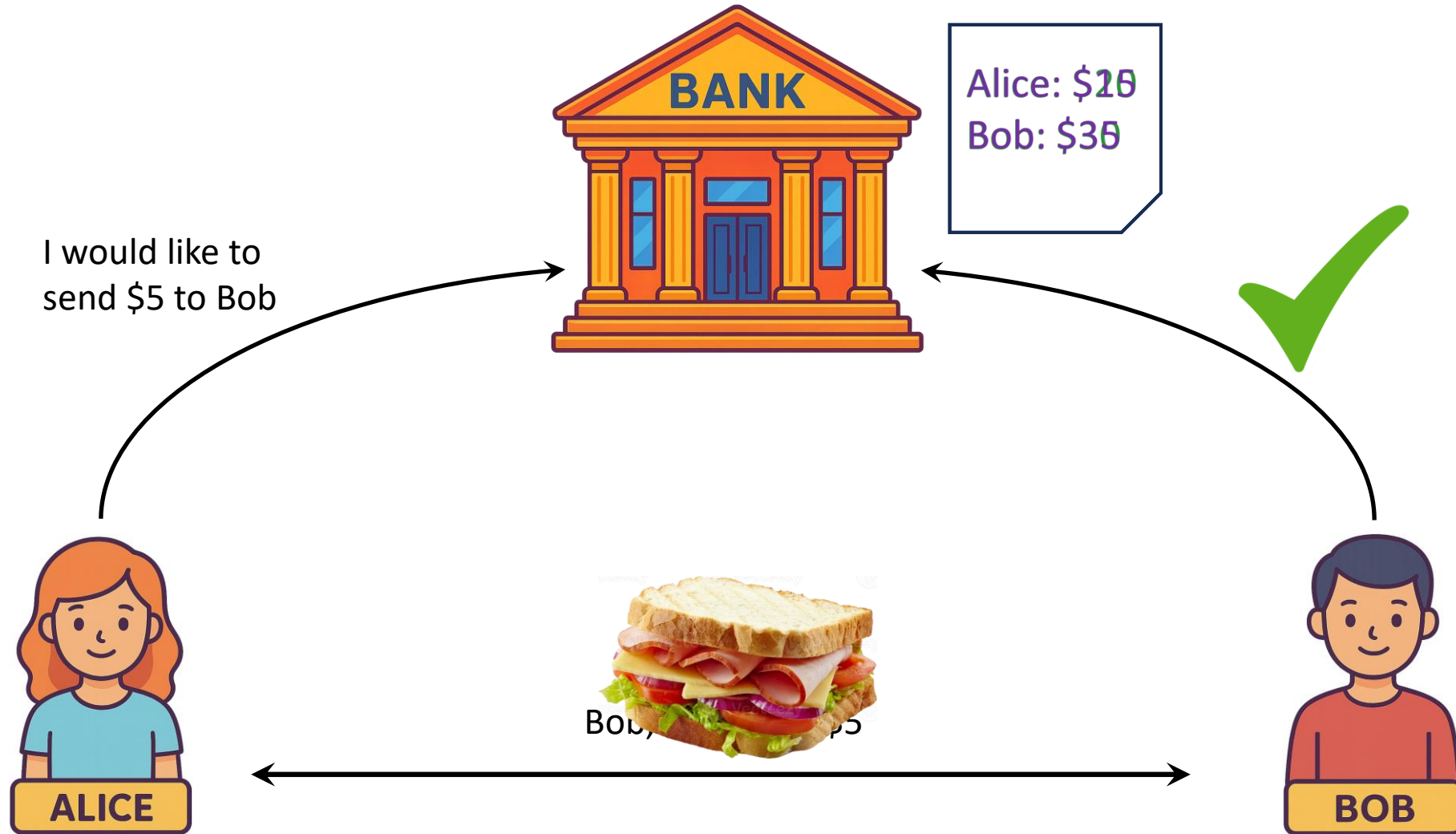
Digital Currency and ePayments

Digital Currency Requirements

- No stealing
- No double-spending
- Minting / creation
- Peer-to-peer payments (direct payments)
- Work without requiring a third party to be online
- Work without an intermediary
- Maintain privacy (at the level of printed currency!)

Let us evaluate if generations of digital money satisfy this

Traditional ePayments



Traditional ePayments

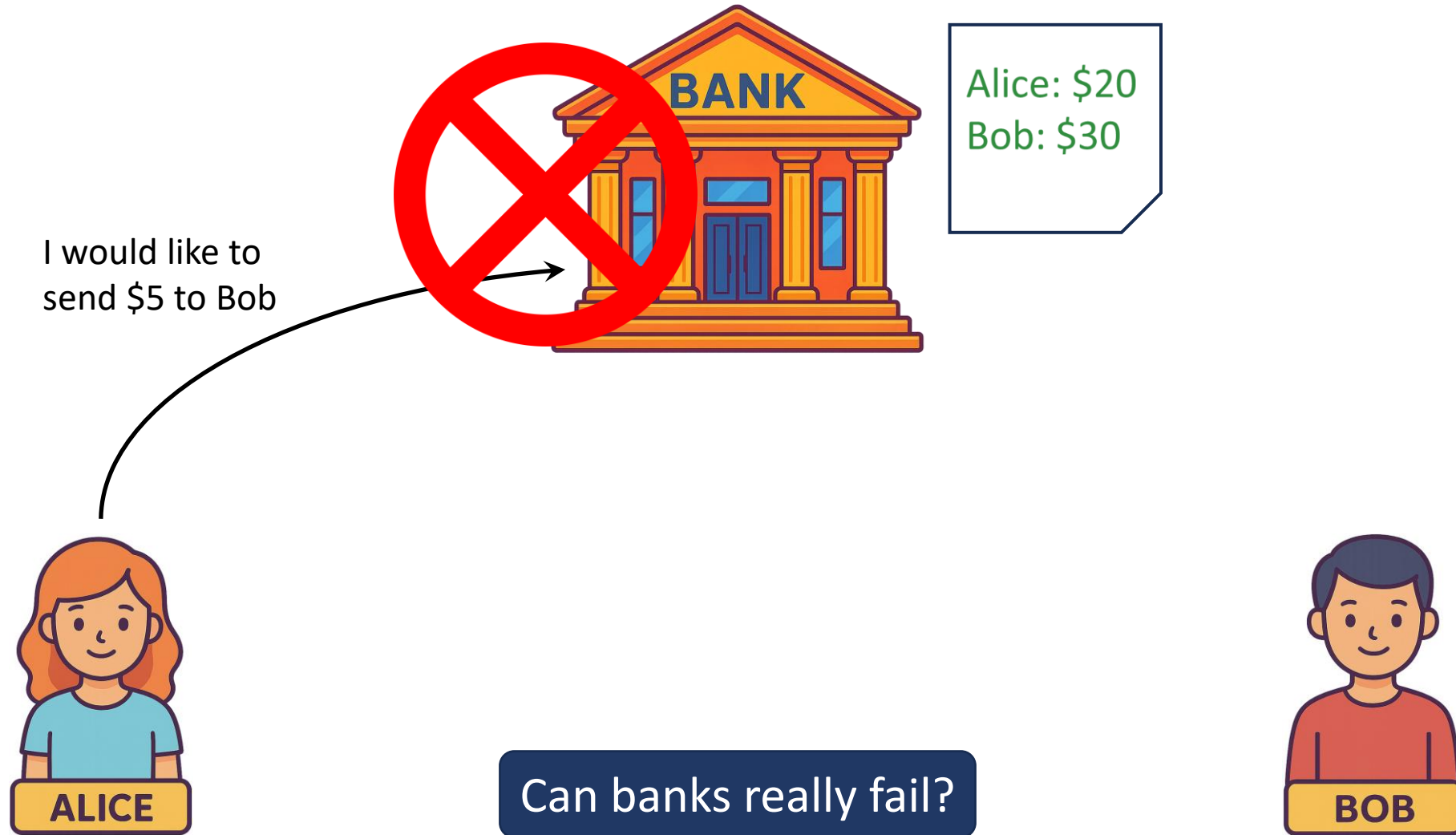
- Pros

- Payment is digital
- Alice and Bob can be at different locations (can they?)

- Cons

- Not peer-to-peer
- Bank can fail
- Bank can delay payments (or censor altogether)
- **Privacy issues** – the bank knows everything
(who is paying how much to whom!)

The banks can fail



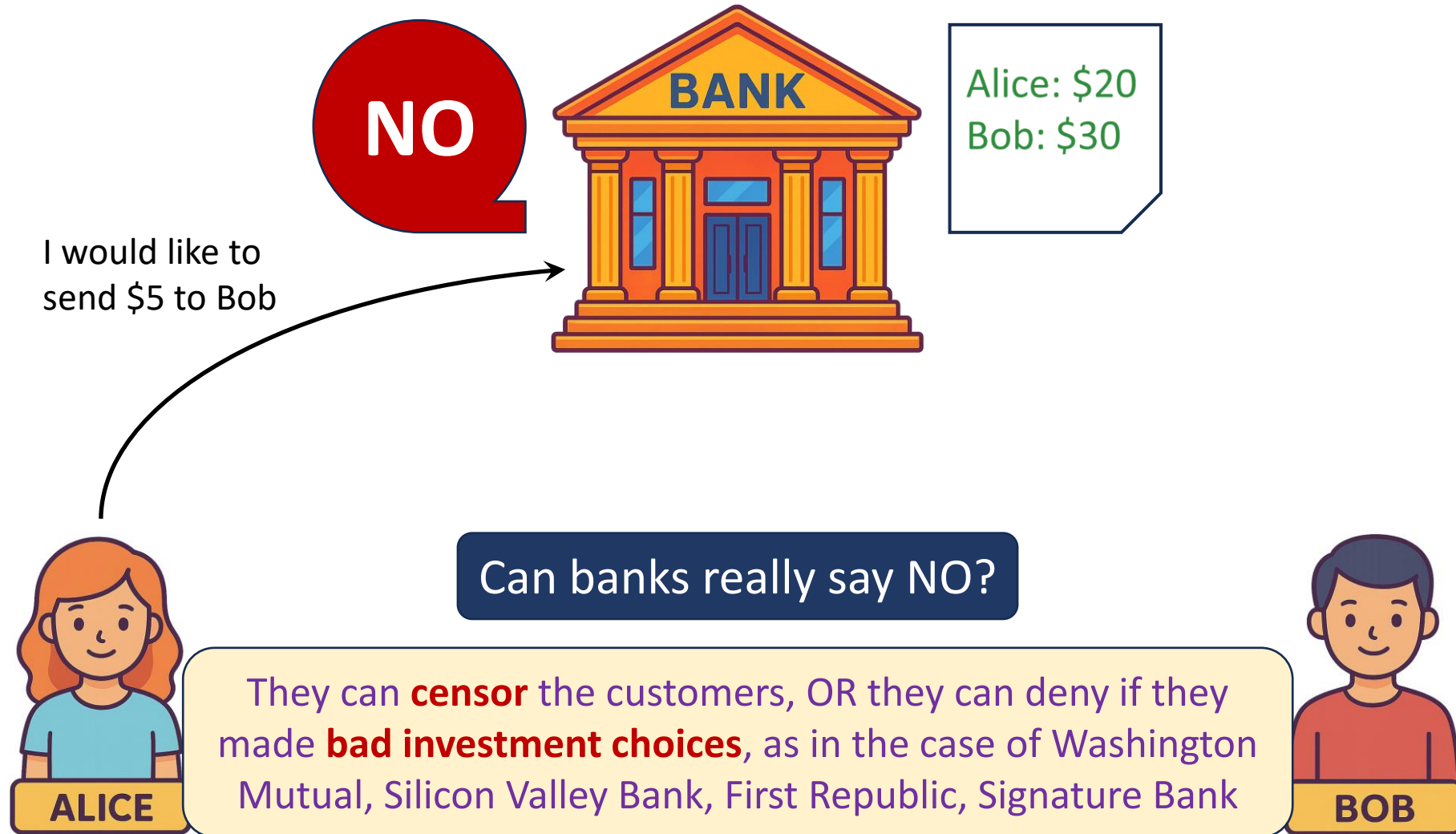
The banks can fail in the real world



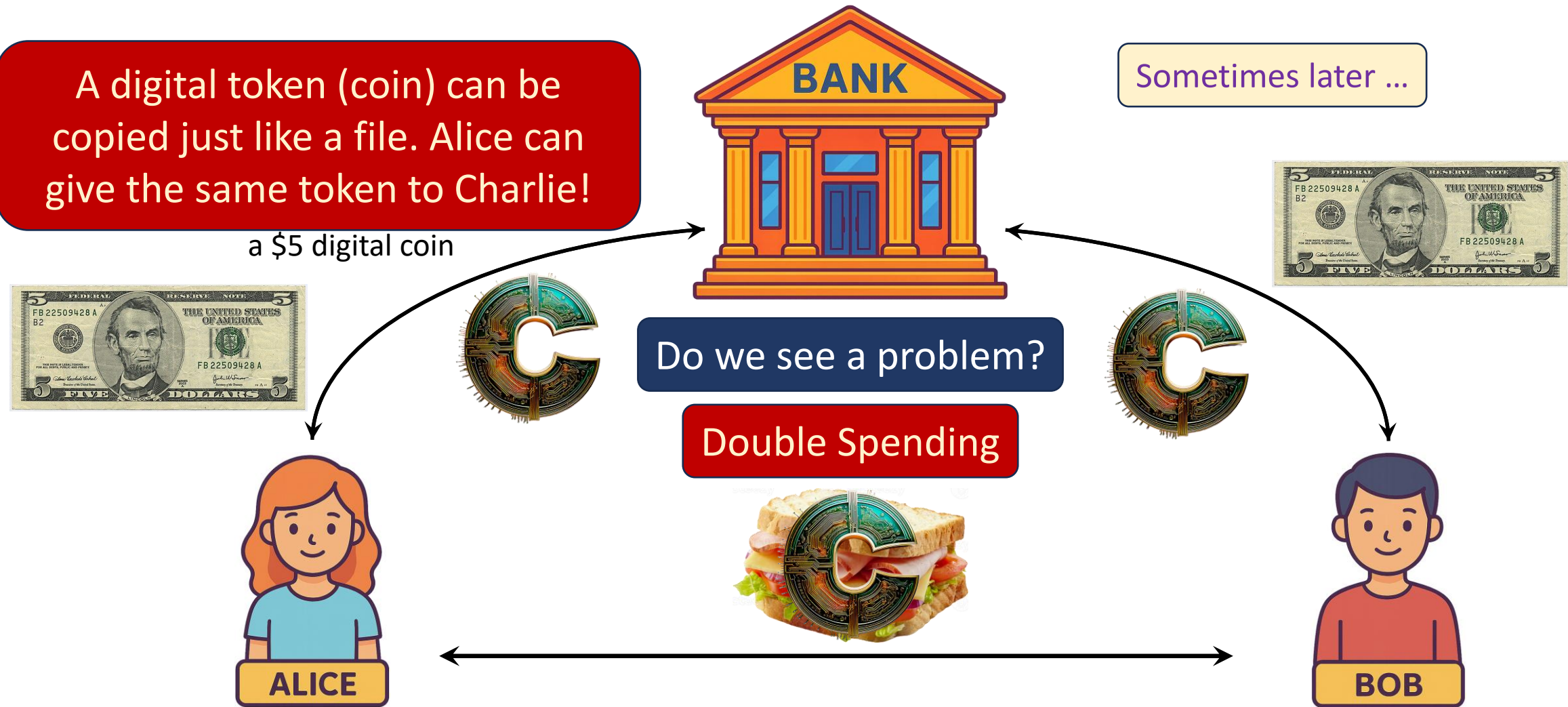
Many Examples

Customers line up at the ATM in Kabul after the US forces left in 08/2021

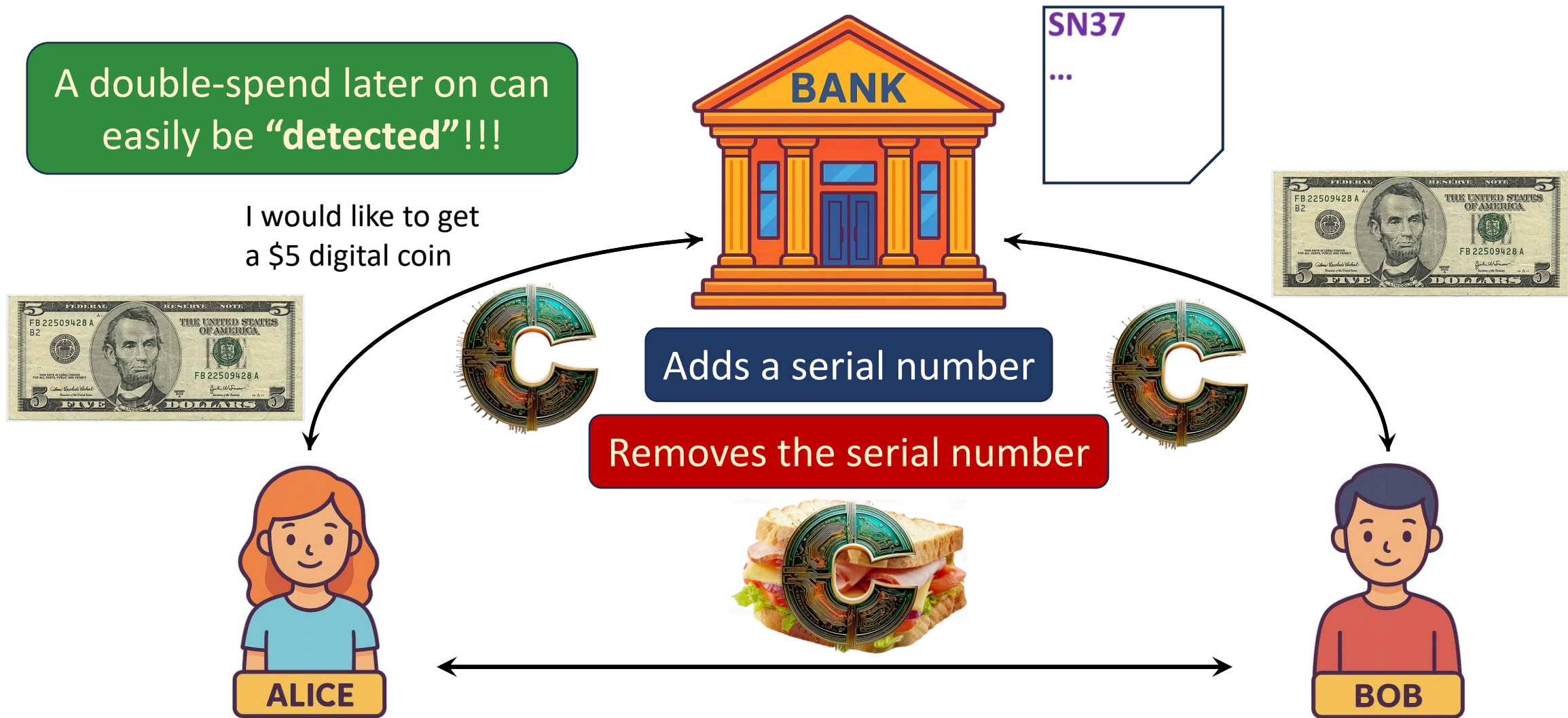
The banks can deny or censor



First Attempt on P2P Payments: Simple E-Cash



Simple E-Cash without double spending



Simple E-Cash

- Pros

- Payment is digital
- Alice and Bob can be at different locations
- Peer to Peer (which is great!)

- Cons

- Full P2P means: double-spending can be detected (but not avoided)
- **Avoiding** double-spending requires the bank to **always** be online
- The bank can fail, delay payments, or censor altogether
- **Privacy:** Can the bank identify the payer, payee, or the payment?
 - Think of the serial number (SN), *which is used to avoid double-spending*
 - Knowing the SN, the bank knows who is paying how much to whom!!!

Chaumian E-Cash using blinded signatures

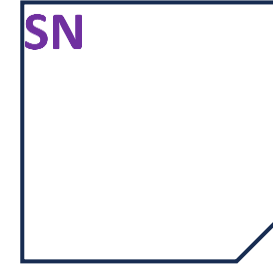
Solves the privacy issue in Simple E-cash

Bank doesn't see the SN

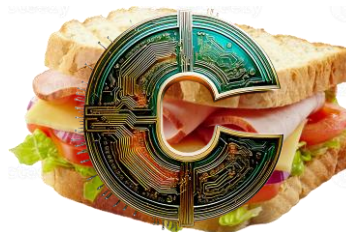
I would like to get a \$5 digital coin and here is $b(SN)$



Bank signs $\text{sig}(b(SN))$

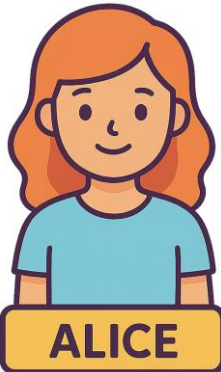


What about double-spending?



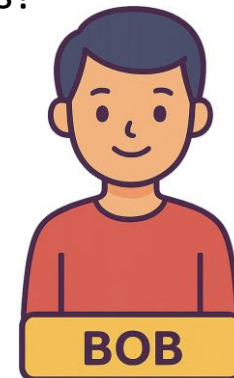
$\text{sig}(SN)$, SN Verifies?

Alice chooses a serial number SN and blinds it



Alice can retrieve $\text{sig}(SN)$

Alice provided $\text{sig}(SN)$, SN to Bob



Chaumian E-Cash

- Pros

- Payment is digital
- Payer and Payee can be at different locations
- Peer to Peer (which is great!)
- **Privacy** is maintained: SN is known to Alice, but not to the bank
- Same privacy guarantee as the physical paper money!

- Cons

- The bank is still the middle party that everyone is forced to trust
- The bank can censor payments

- **What about double-spending?**

- Detection and identification possible (only after the fact!)

Recap: Money

- Barter, Representative, Fiat
- Banks
 - Always are in-the-way
- Simple E-cash
 - Privacy is violated
- Chaumian e-cash
 - Maintains privacy (via blinding factor)
 - Can even avoid deter from double spending
- Getting rid of the central entity (bank) requires decentralization
 - Case for blockchain? (leads to cryptocurrency)

Acceptability of Chaumian E-Cash

- Why did Chaumian E-Cash not fly commercially?
- It was considered too private
 - Anon digital payments lead to tax evasion, money laundering, etc.
- Online verification required: poor user experience
 - Over a possibly unreliable or unavailable (1990s) network
 - Cash and credit cards remained the champions!
- Business Model issues: no incentive for banks
 - Less control to banks, more privacy to customers

Too private for the government, too inconvenient for users,
too unattractive for banks

Desirable Features of Money

- What do we need the money to be?
- Durable
- Portable
- Divisible
- Fungible
- Acceptable
- Stable – limited Supply – hard to counterfeit
- Are these features desirable for Non-digital or digital money?

Early Digital Currencies: Shortcomings

Why did the earlier digital currencies fail? *In other words:
What shortcomings of digital payments did Bitcoin address?*

- Is it Peer-to-peer?
- Does it avoid stealing?
- Is it widely accepted by the merchants?
- Is there a well-defined process for minting/creation?
- Does it offer an agreeable form of a record book/ledger?
- Does it allow the transacting parties to remain private?
- Does it avoid double-spending?

(Crypto)currency Challenges

- **No stealing**
 - Only Alice can move her money. Solution?
 - Cryptographic digital signatures
 - Who runs the PKI?
- **No double-spending**
 - Alice can not duplicate her money. Solution?
 - Global ledger
 - Who maintains the public view?
- **Minting**
 - Fair money creation. Solution?
 - Proof of Work
 - Who hands out money for solving puzzles?

CBDC, Bitcoin, Chaumian E-cash

Feature	Chaumian e-cash	Bitcoin	CBDCs
Introduced	1980s–1990s (DigiCash by David Chaum)	2009 (Satoshi Nakamoto)	2020s (various pilots: China, EU, etc.)
Issuer	Central bank or licensed bank	No issuer (decentralized network)	Central bank
Privacy	✅ Strong — blind signatures unlink withdrawal from spending (cash-like anonymity)	⚖️ Pseudonymous — transactions are public, but identities can be hidden unless addresses are deanonymized	❌ None — transactions are fully traceable by the central bank
Double spending solution	Bank's spent-coin database + cheater-revealing protocol	Public blockchain consensus (proof-of-work, miners)	Central bank's centralized ledger
Centralization	Centralized (depends on a single issuer)	Decentralized (peer-to-peer nodes)	Centralized (one official ledger)
Government stance	🚫 Rejected — too much privacy, undermines AML/KYC	⚖️ Tolerated but regulated — transparent enough for surveillance via exchanges	✅ Actively promoted — governments want control & oversight
Adoption outcome	Failed commercially (DigiCash bankrupt in 1998)	Global adoption, large ecosystem	Pilots and rollouts ongoing (e-CNY, digital euro, etc.)
Main strength	Cash-like anonymity online	Decentralization, censorship resistance	Efficiency, surveillance, policy control
Main weakness	Needed banks to cooperate; governments disliked its privacy	Price volatility, scalability, partial privacy	Raises privacy concerns, potential for abuse of control

Lecture 08

Thursday, Feb 12, 2026

Consensus in Blockchains

- **What is consensus?**

- Reaching an agreement in a group
- Blockchain requires it to be done dynamically, on-the-fly
- **Practical constraint:** Can not take forever to reach an agreement

- **Why is consensus needed?**

- Avoids conflicting transactions T1 and T2
- Say T1 and T2 are conflicting (double spending, for example)
- Assume that T1 and T2 are in separate journals
- What assures that T1 and T2 can't be part of the same ledger?

- ***Dishonest nodes may not reach consensus, and that's okay!***

Consensus via Majority

- How is consensus reached
 - For two nodes: if both are honest
 - For many nodes: ???
- Unified and correct view (ledger) evolves:
 - If the majority of the nodes are honest
 - Double-spending (conflicting transactions, in general) is avoided
- What is “Majority”?
 - An individual?
 - A computer?
 - What about one user creating multiple (dishonest) nodes?
 - A weighted majority is considered – each node is assigned a weight

Proof of Work (PoW) Consensus

- **Roots in SPAM filtering**
 - A moderately expensive, but not intractable, puzzle to solve
 - No impact on good users, but disincentivizes spammers
- **How is it used in Blockchain?**
 - A node that solves a computationally intensive puzzle first *wins*
 - It is telling the truth (i.e., all transactions are valid)
 - The result of the computation is easy to verify by other nodes
 - The block of the winner node is added to the blockchain
- **Food for thought**
 - What if the “work” is minimal (or none!) in a PoW blockchain?

Proof of Stake (PoS) Consensus

- Designed to address high energy consumption in PoW
 - Initially, both Bitcoin (BTC) and Ethereum (ETH) used PoW
 - ETH switched from PoW to PoS
- Process (specific to Ethereum)
 - Validators (miners!) “stake” their ETH to participate
 - The system chooses a random validator node to create new blocks
 - A node with a higher stake is more likely to be chosen
 - The chosen validator shares the block with the network
 - No heavy computation is needed – staking is required
 - If the node is found to be cheating, it will lose its security deposit
 - 51% or more of staked ETH is needed to defraud the system

Other Consensus Mechanisms

- Proof of Activity
 - A hybrid of PoW (for header only) and PoS
- Proof of Storage
 - Node importance is proportional to the storage capacity
- And many others: Proof of ...
 - Authority, Burn, Capacity, Assignment, Transfer, Elapsed Time
- Practical Byzantine Fault Tolerance (PBFT)
 - A 3-phase protocol
 - Consensus reached if more than $2/3$ nodes are honest
 - Used when the number of participating nodes is known

Popular and Recent Topics Smart Contracts, Stablecoins, and Tokenization

Smart Contracts

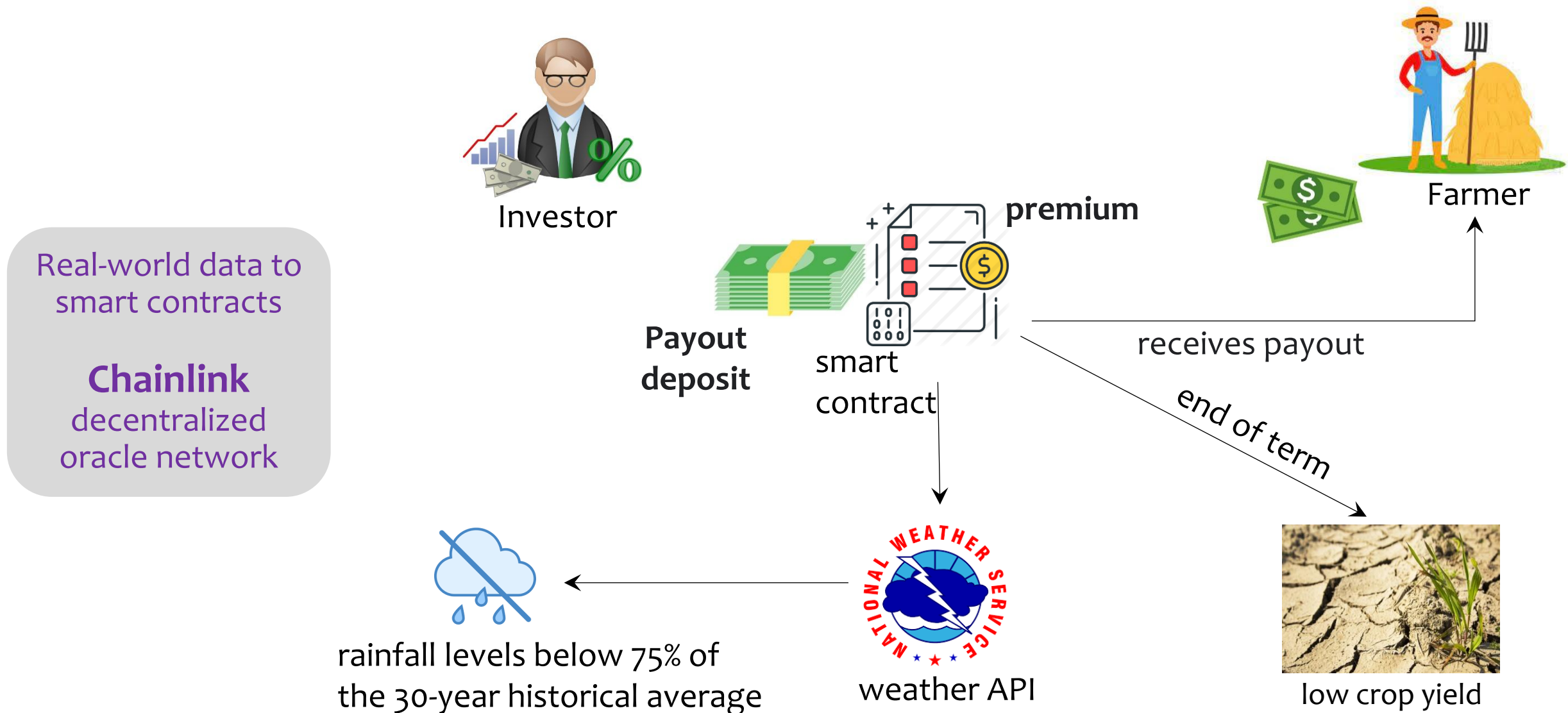
- A program that runs on the blockchain
 - Encodes a self-executing agreement based on predefined events
 - **Settlements** in a smart contract are automatically triggered
 - When the pre-agreed conditions in the contract are met

Result: Automatic Transactions without intermediary

- **Example:** *Crop insurance for weather risk protection*
 - **Typical terms:** if rainfall is below a certain level, payout is made
 - **Farmer(s):** premium (in crypto) locked in the smart contract
 - **Investor(s):** payout (also in crypto) locked in the same contract

Rainfall determined externally; instant payment transfers

Smart Contracts: Example



[Digital] Assets Across Multiple Systems

- **Trend:** Representing assets of value in different systems
- Why needed?
 - Some things can be done better in the new system
- Examples
 - Why wBTC for BTC?
 - Why create USDC pegged to USD?
 - Why tokenize the stock of Apple?
 - Why tokenize the Roosevelt Hotel?
 - Why tokenize gold?

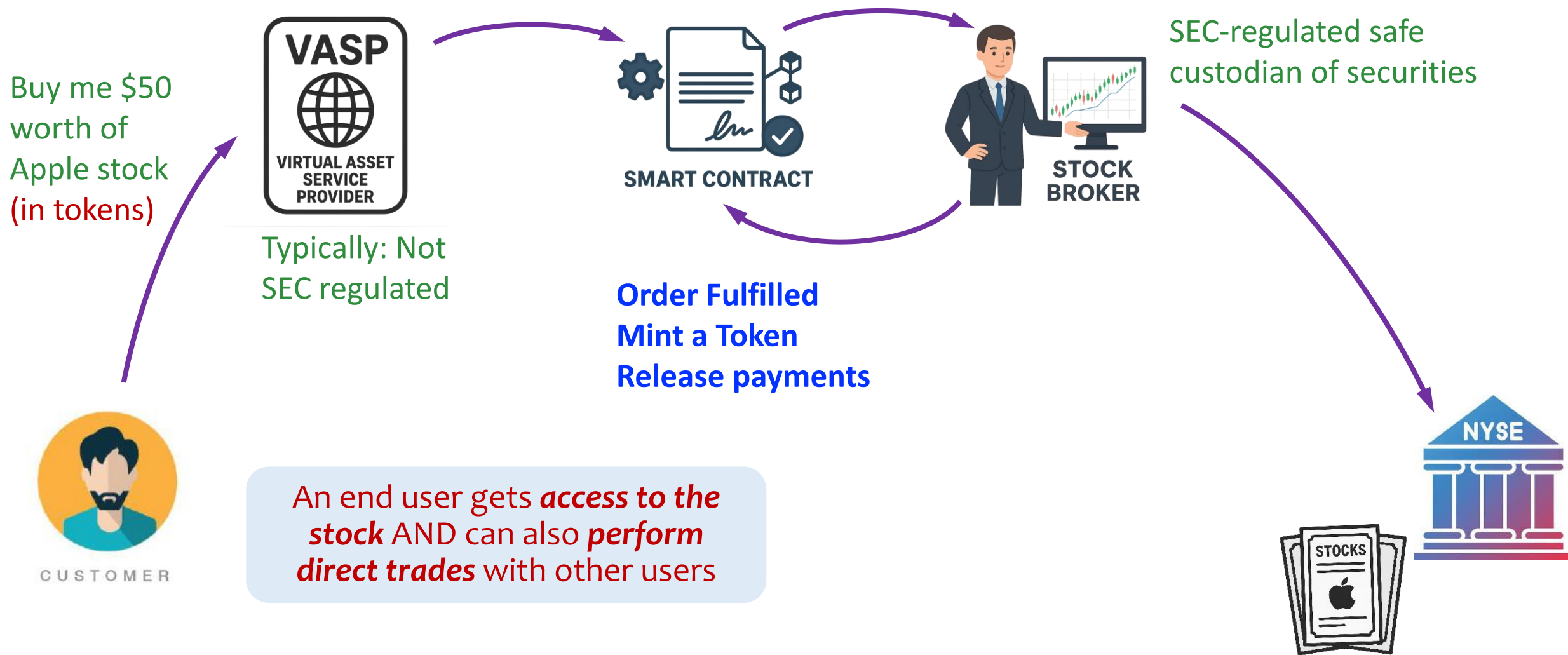
[Digital] Assets Across Multiple Systems

- **Trend:** Representing assets of value in different systems
- Why needed?
 - Some things can be done better in the new system
- Examples
 - Why wBTC for BTC? Unlock huge value and use programmatically
 - Why create USDC pegged to USD? P2P transactions, fast and cheap
 - Why tokenize the stock of Apple? Access to those who don't have it
 - Why tokenize the Roosevelt Hotel? Allow fractional ownership
 - Why tokenize gold? Convenience in gold trading

Representing digital assets across systems

- Does a stock exchange do the same?
- Does a stock exchange bring transfer of value across systems?
 - Value of the company represented as stocks
 - Purpose: Public fundraising
- What does tokenization of stocks bring to the table?
 - Trading and transactions made P2P
- Can the stock exchange be bypassed?

Example: Buying Tokenized Stock



Layer 2

- What is Layer 2?
 - Secondary protocol built on top of an existing blockchain system
- Main goal(s) of Layer 2?
 - Provision of higher transaction speeds
 - Possibly bringing in functionality that the mainchain may not offer
- Examples of Layer 2?
 - Stacks for Bitcoin
 - Polygon for Ethereum

CBDC

- A government-issued stablecoin
 - Recall: Stablecoin is a token (modern-day representative money)
- Noting: Most fiat is NOT printed (only a small percentage!)
 - It is already “digital”
- Why then use CBDC instead of digital fiat?
 - CBDC is governed by the central bank
 - Digital fiat by commercial banks